Models of the motor system

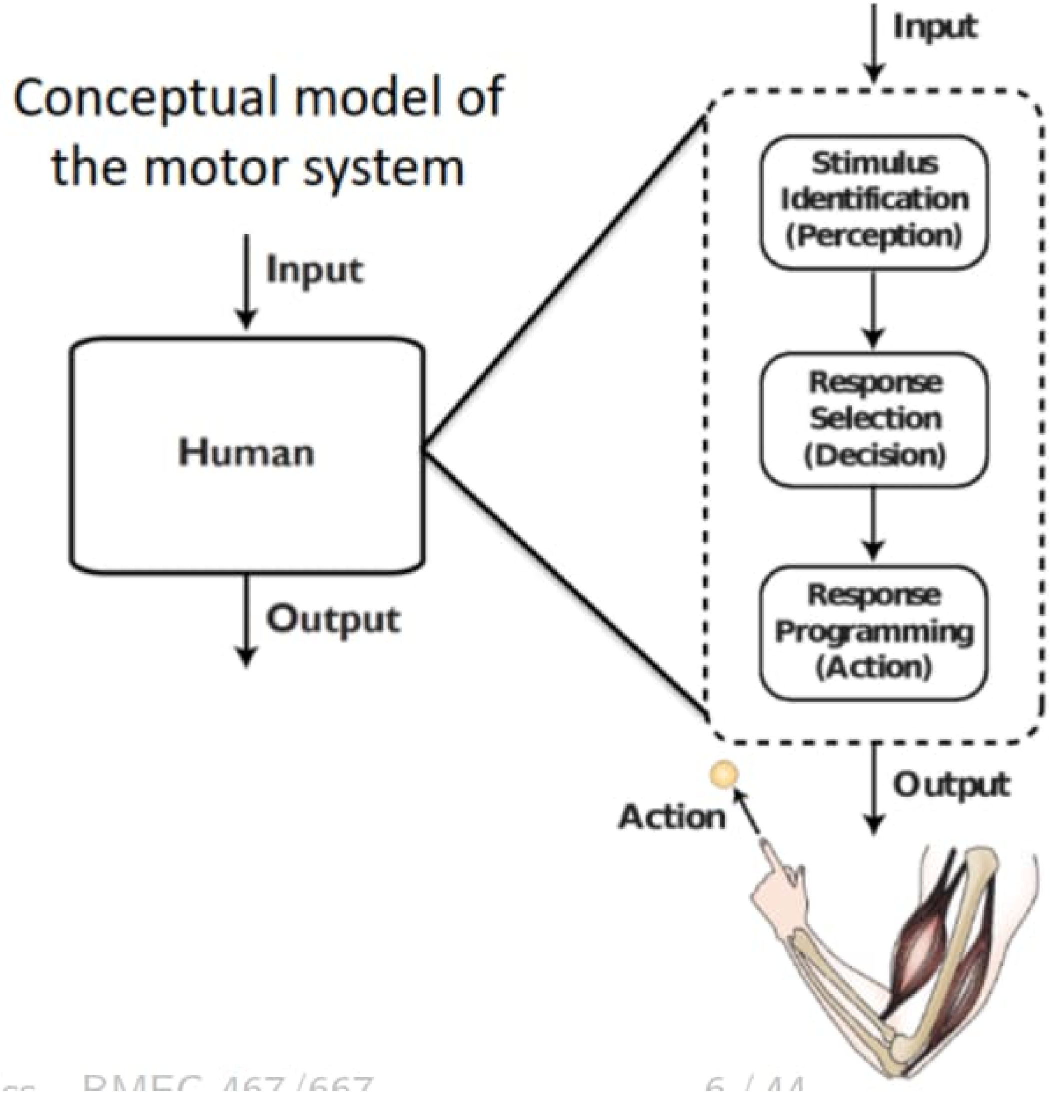
Spring 2025

Opher Donchin

Part 1

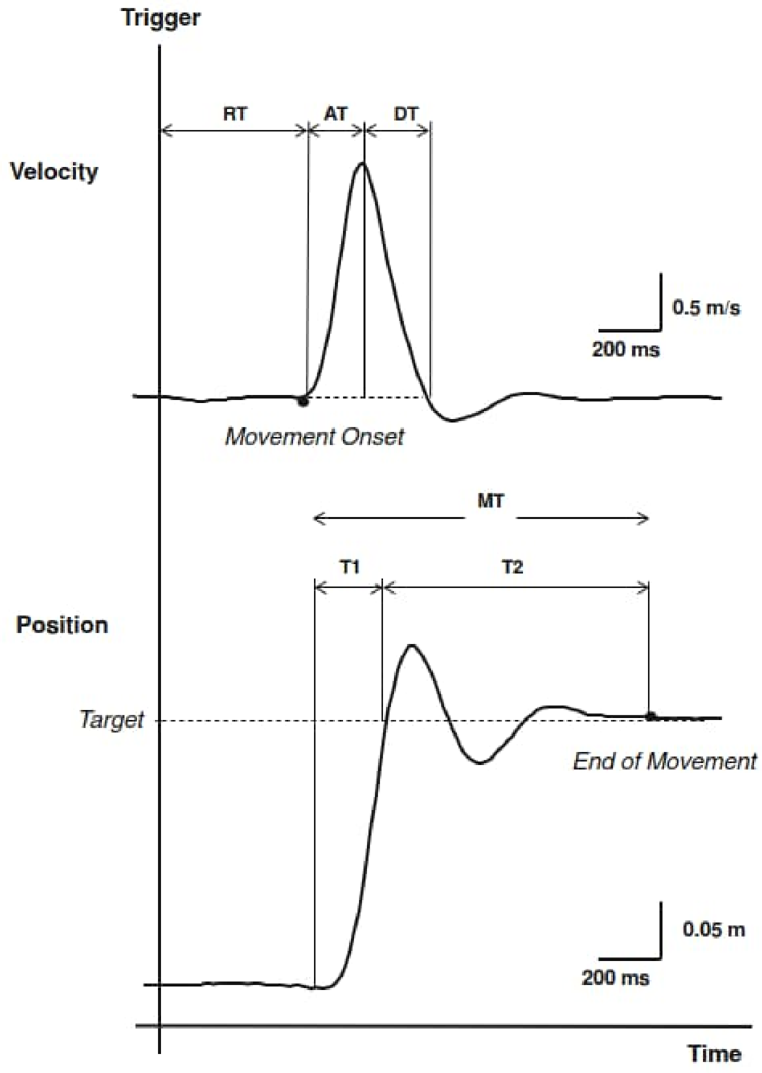
- Hick’s law

Feedforward motor processing



Reaction time

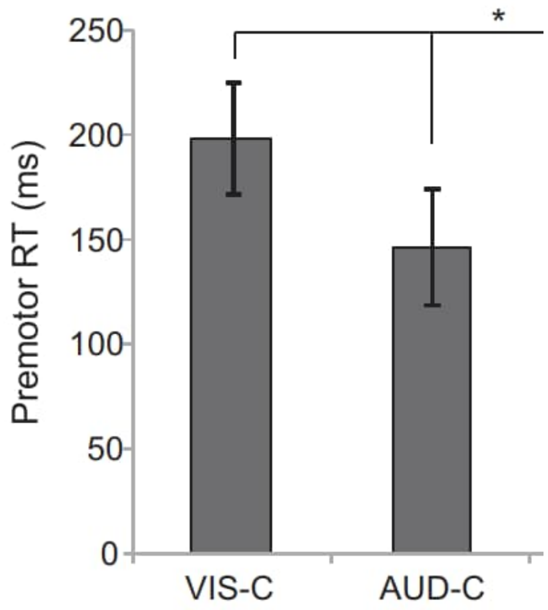
- Factors that influence reaction time
  - Number of choices
  - Stimulus mode and intensity
  - Stimulus-response compatibility
  - Anticipation



Di Lorenzo *EBR* 2012 doi:10.1007/s00221-012-3318-y

Reaction time

- Factors that influence reaction time
  - Number of choices
  - Stimulus mode and intensity
  - Stimulus-response compatibility (demo)
  - Anticipation



Carlsen *Neuroscience Letters* 2011 doi:10.1016/j.neulet.2011.06.027

Reaction time

- Factors that influence reaction time
  - Number of choices
  - Stimulus mode and intensity
  - Stimulus-response compatibility

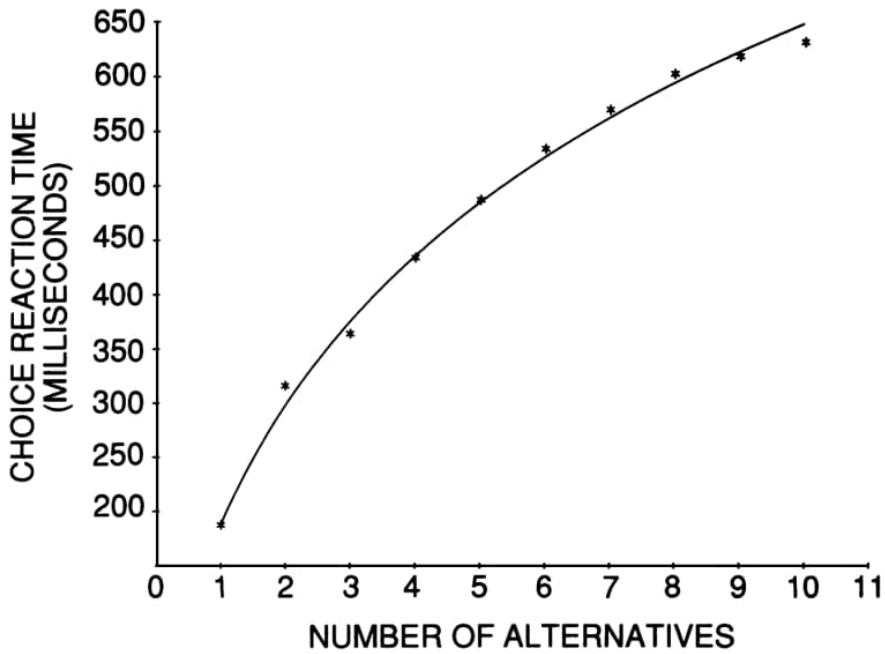
Demo

Di Lorenzo *EBR* 2012 doi:10.1007/s00221-012-3318-y

Reaction time

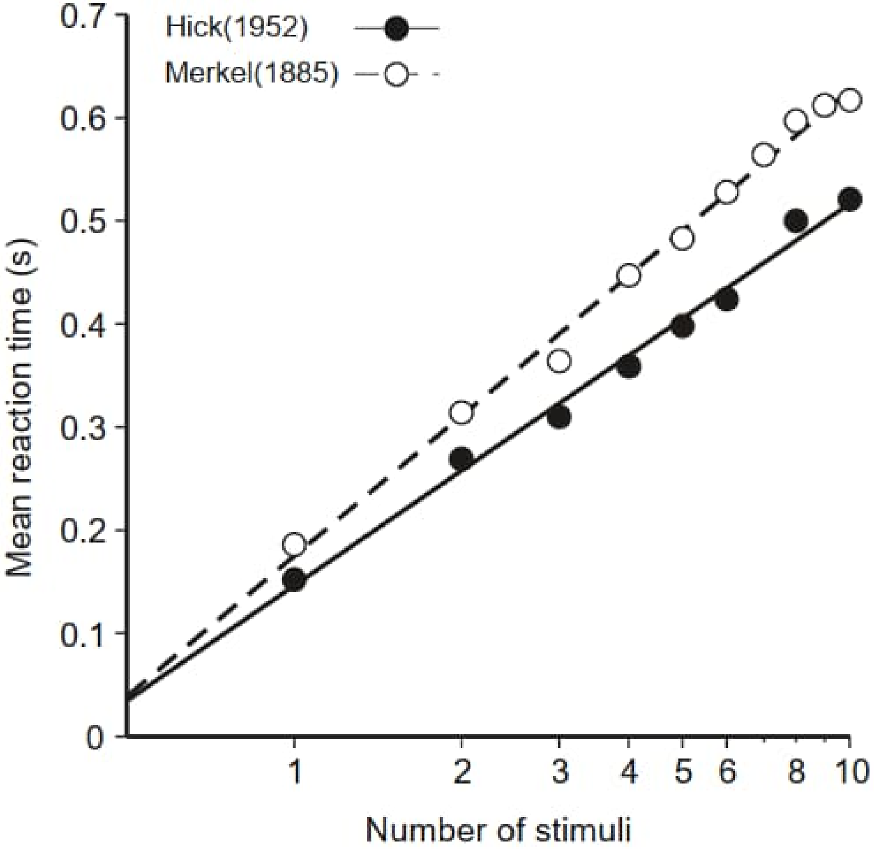
- Factors that influence reaction time
  - Number of choices
  - Stimulus mode and intensity
  - Stimulus-response compatibility (demo)
  - Anticipation

Reaction time increases with number of choices



Data from Merkel PhD Thesis 1885
via: Norwich 2003 Chapter 13 http://www.biopsychology.org/norwich/isp/isp.htm

Hick’s law



$$RT = a + b\log_2(N)$$

a: RT for a single choice

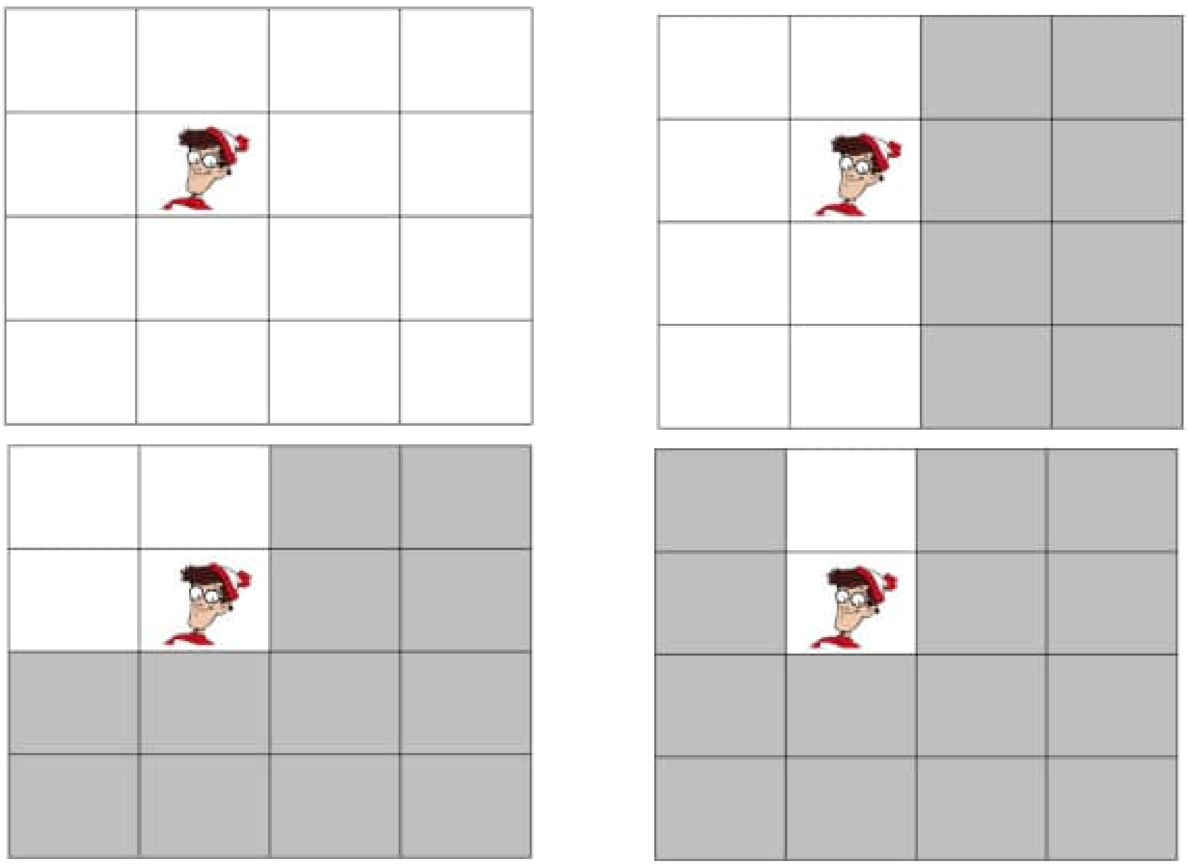
b: Increased RT for doubling the number of choices

N: Number of choices

Laming *Entropy* 2010 doi:10.3390/e12040720

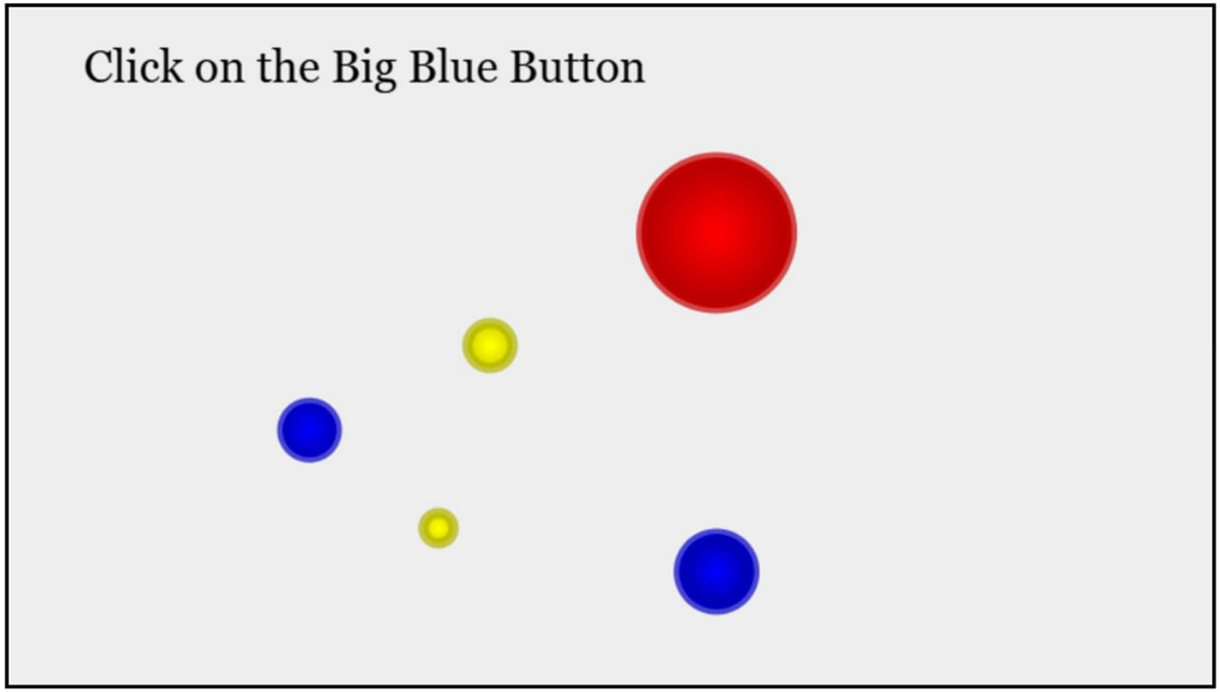
Why $\log_2 N$?

- Binary search



Online Hick's law demo

- Test the law for yourself
- https://cpe-iitg.vlabs.ac.in/exp/hick-hymans-law/simulation.html



No of Buttons:	Time(in secs)
Starting:	1.891
1:	0.274
2:	1.631
5:	1.885
8:	3.807
8:	2.876
9:	2.158
5:	1.869
9:	2.556
9:	2.034
10:	2.003
:	undefined

Part 2

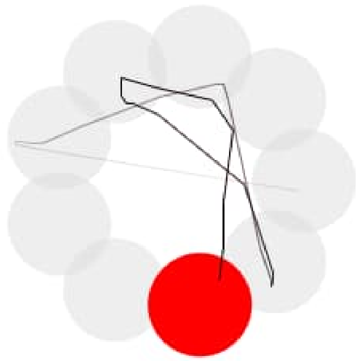
- Fitt’s law

Fitt's law

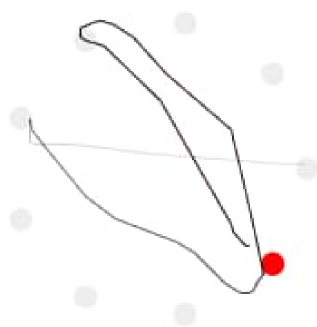
- Movement time scales with difficulty
  - Distance
  - Precision

Fitt's law

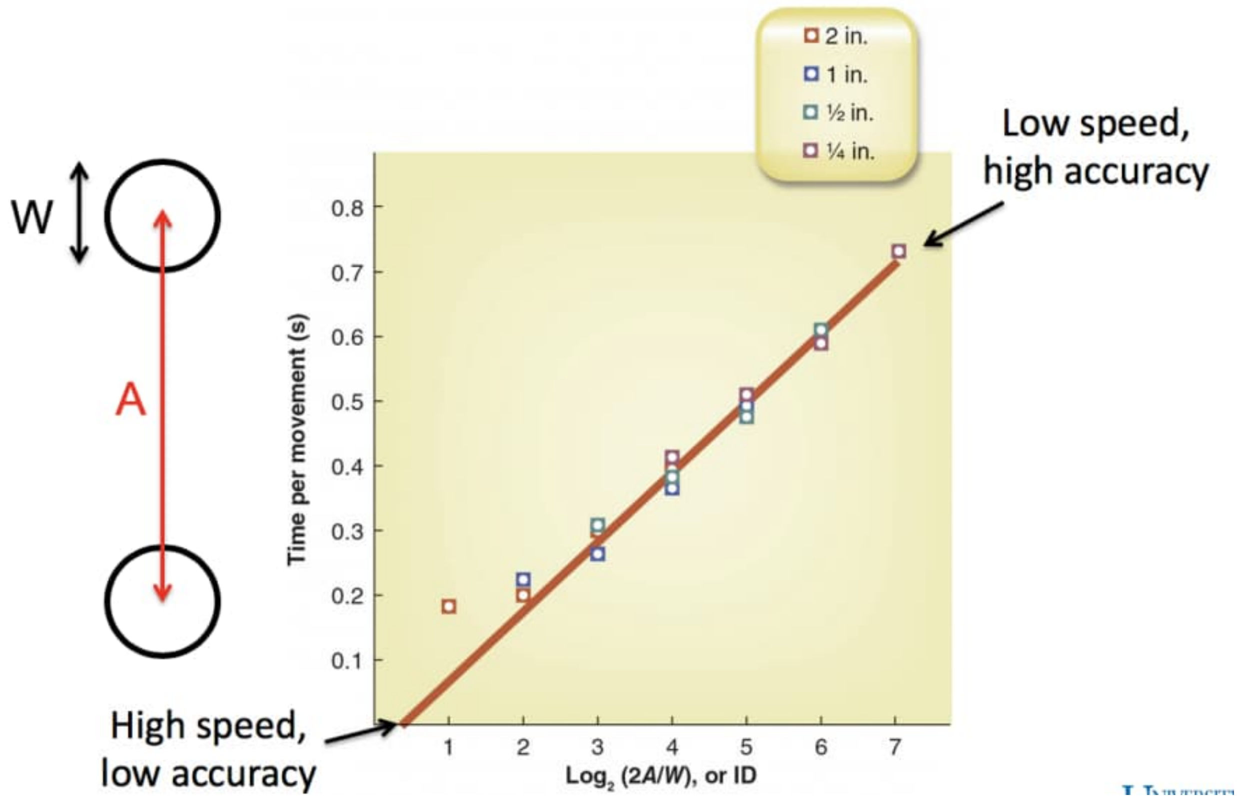
- Movement time scales with difficulty
  - Distance
  - Precision

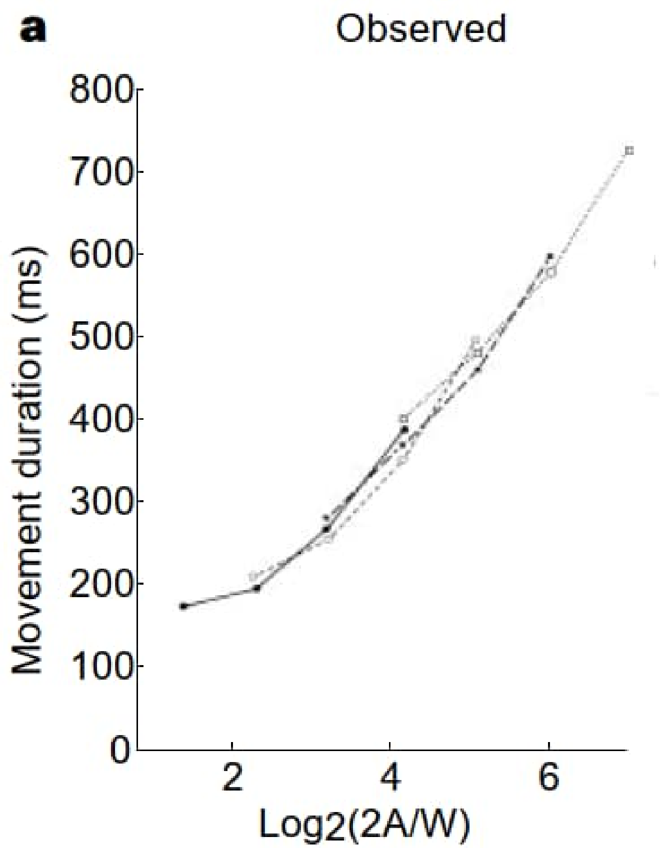


$$MT = a + b\left(\log_2\left(2\bullet\frac{\text{target distance}}{\text{target size}}\right)\right)$$



Original Fitt's data

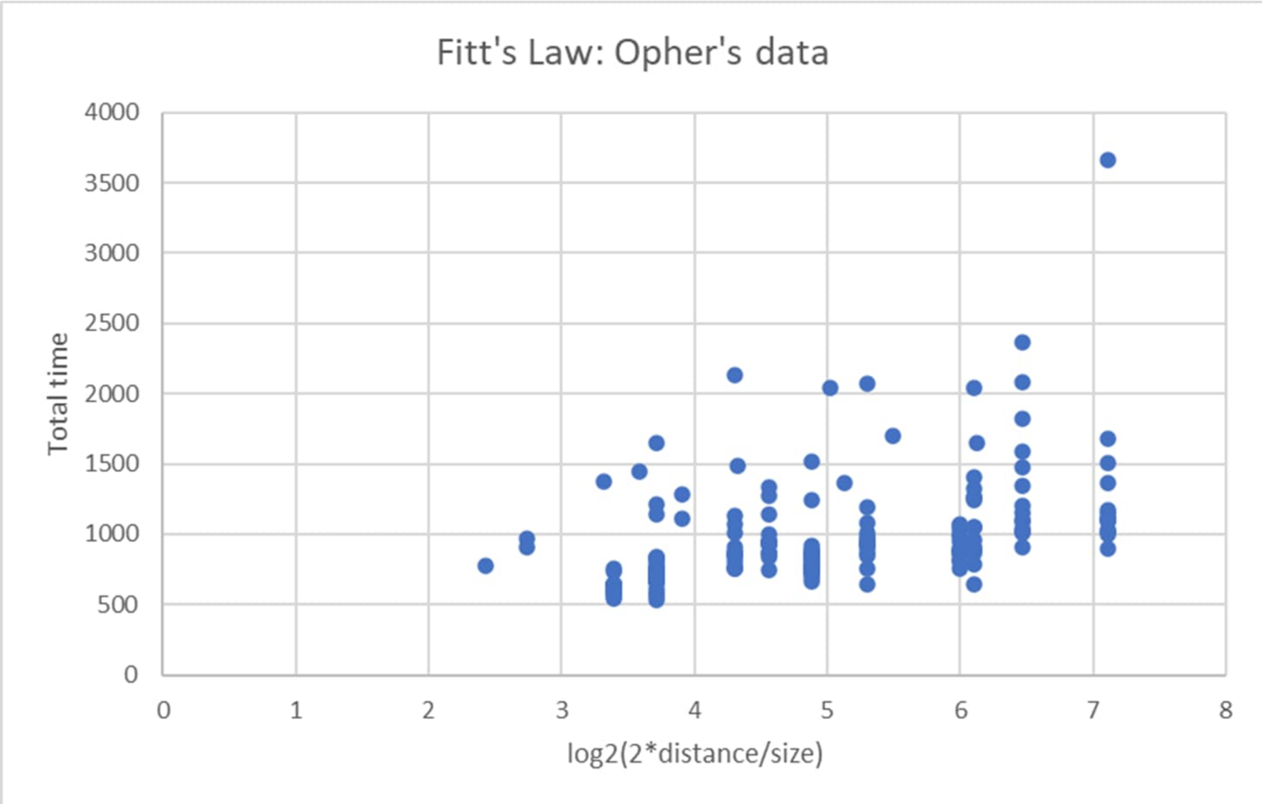




Fitts *J Exp Psych* 1954 doi:10.1037/h0055392
via: Schmidt 2011 ISBN: 9781492547754
Harris *Nature* 1998 doi:10.1038/29528

Try Fitt's law on your own

- Many online experiments to try
- https://twinji.github.io/fitts-law-input-task/#/



Fitt’s law used in web design

- Frequently used areas
  - Large and central
  - Reduce time and errors
- Infrequently used areas
  - Small and peripheral
  - Increased accuracy requirement:
    - Prevents errors
    - Costs time

Part 3

- The minimal variance principle

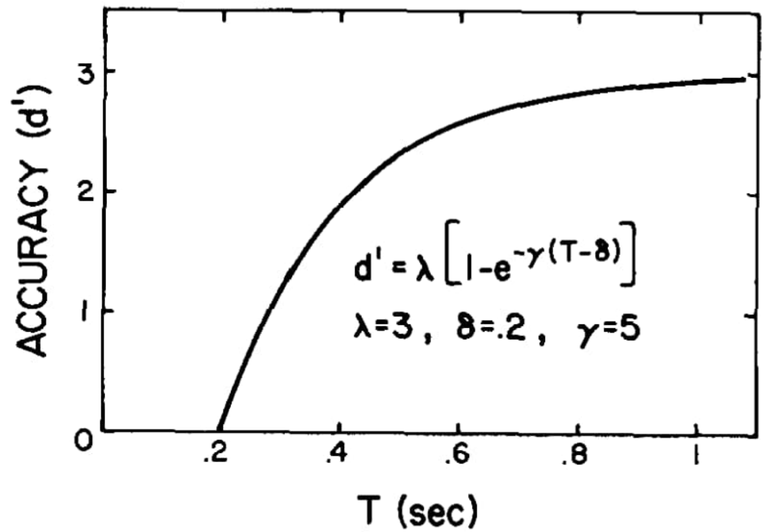
The speed / accuracy tradeoff

- Fix distance
- Call 1/target size the accuracy

$$MT = a + b\left(\log_2\left(2\bullet\frac{\text{target distance}}{\text{target size}}\right)\right)$$

$$MT = \left(a + b\log_2 2\left(\text{target distance}\right)\right) + b\log_2\left(\text{accuracy}\right)$$

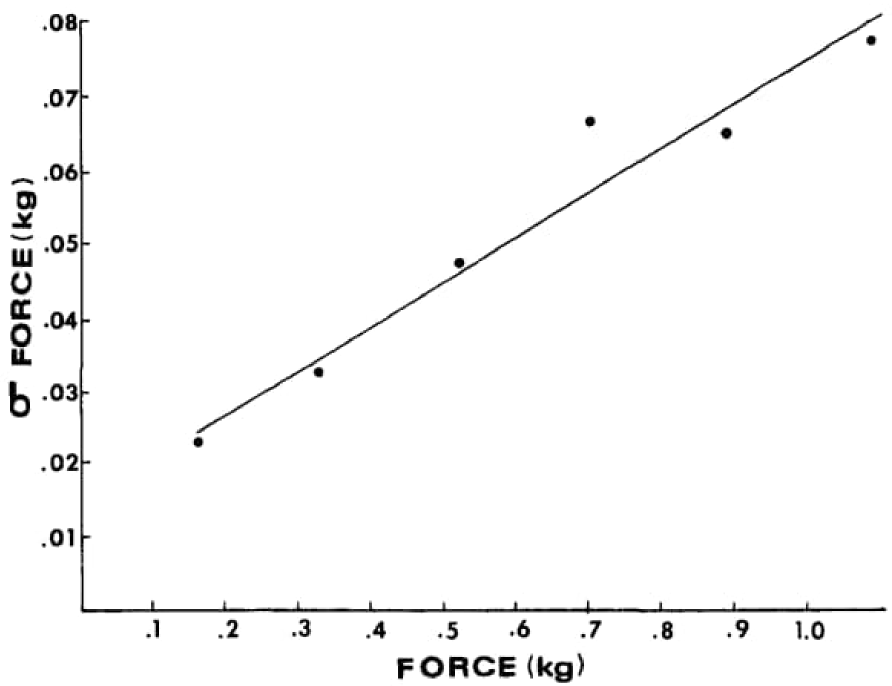
$$= a' + b\log_2\left(\text{accuracy}\right)$$

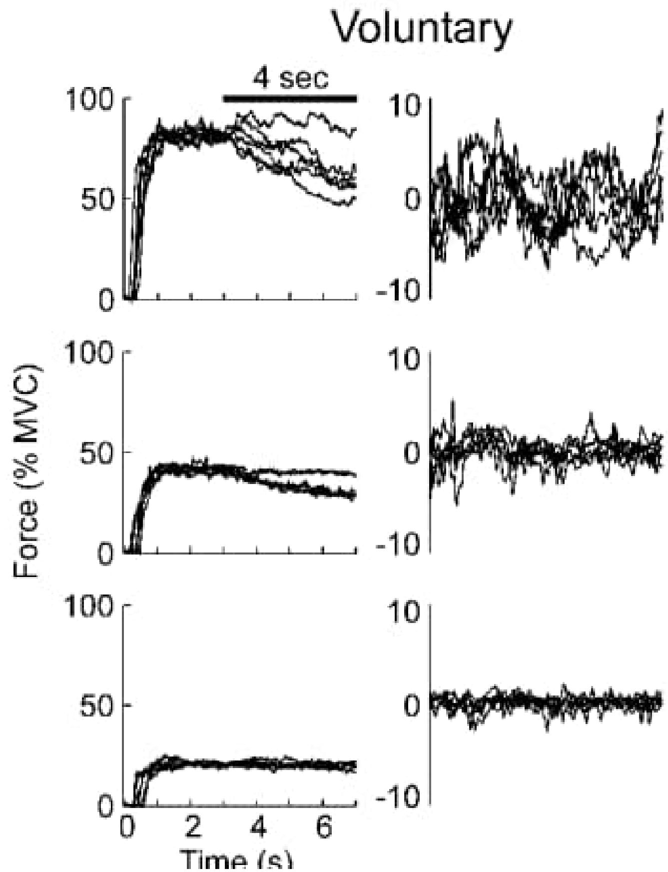


Wickelgren *Acta Psychologica* 1977 doi:10.1016/0001-6918(77)90012-9

Motor variability scales with motor output

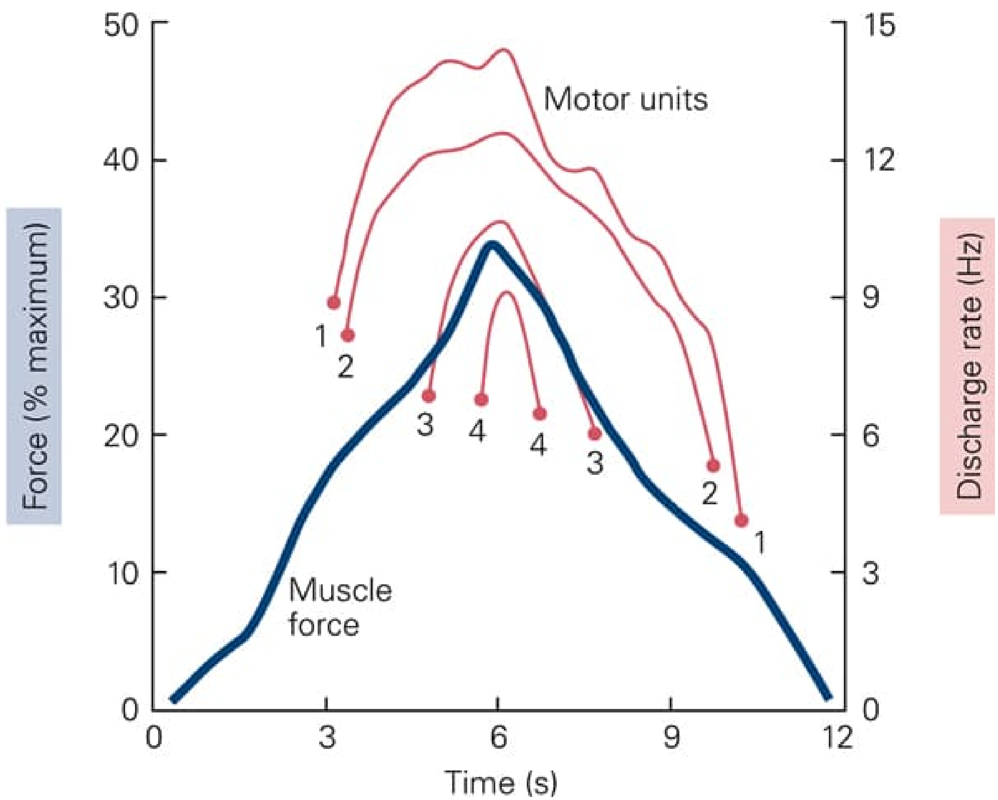
- Signal dependent noise





Schmidt *Psychological Review* 1979 doi:10.1037/0033-295X.86.5.415
Jones *J Neurophysiol* 2002 doi:10.1152/jn.00985.2001

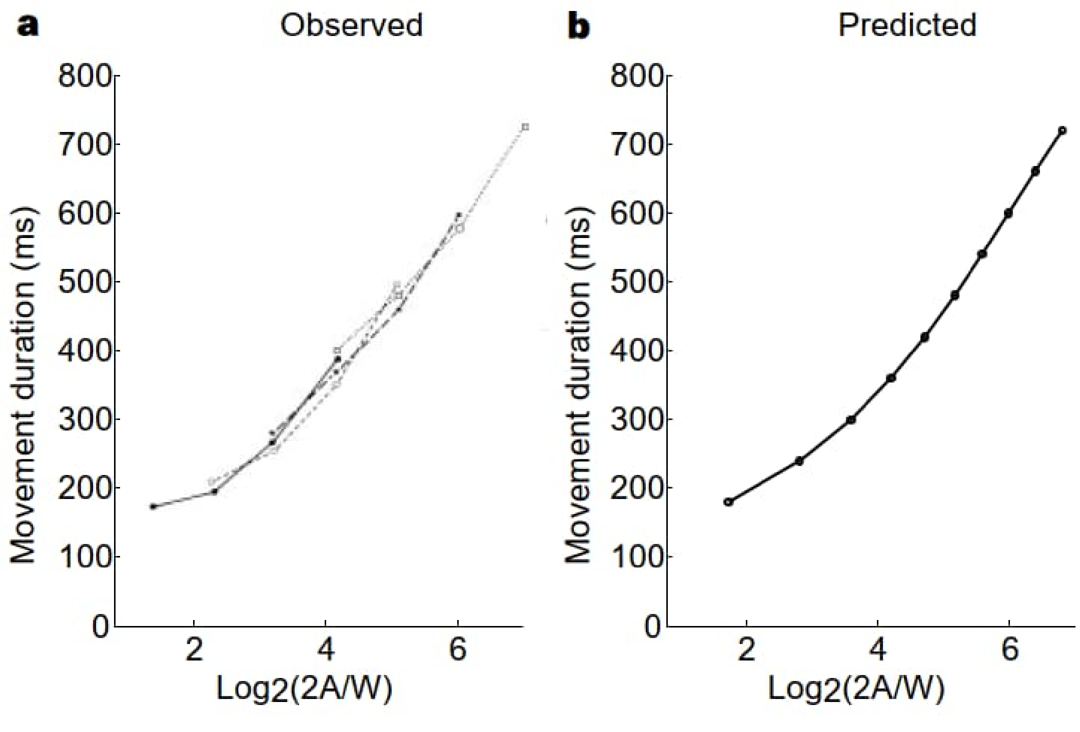
Size principle explains signal dependent noise



- Increased force recruits bigger motor units
- Bigger motor units have bigger fluctuation

Model: minimize variability

- The brain chooses motor commands to minimize motor variability
  - Given time constraints



Harris *Nature* 1998 doi:10.1038/29528

Model of 1-D arm movement

- Signal dependent noise

$$x_{t+1} = Ax_t + B\left(u_t + w_t\right)$$

x_t: state (position, velocity)

u_t: motor command

w_t: noise $N\left(0, ku_t^2\right)$

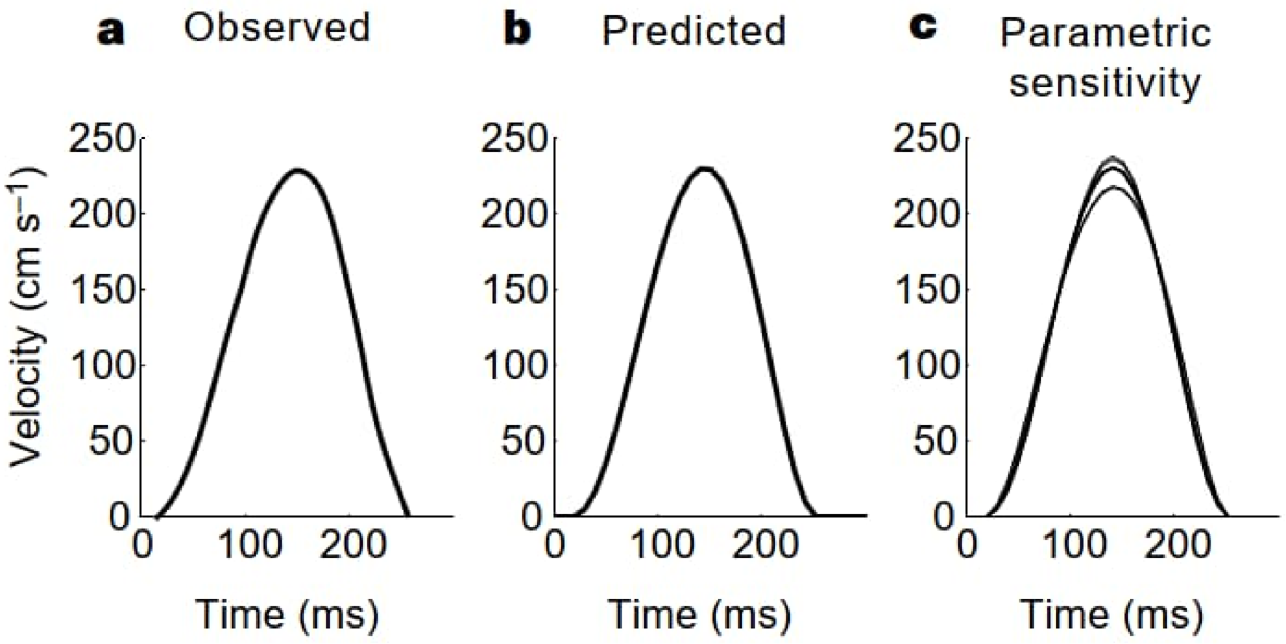
Iterate to make a movement

- Constrain to reach target at given distance and time
- Find u_t to minimize variance

Now find minimum time to achieve given accuracy constraint

Harris *Nature* 1998 doi:10.1038/29528

Minimum variance also gives bell shaped profile



Harris *Nature* 1998 doi:10.1038/29528

Part 4

- The minimum jerk trajectory

Minimum jerk is an older idea

- Assume movement should be smooth

$$C = \frac{1}{2}\int_0^T \left(\left(\frac{d^3x}{dt^3}\right)^2 + \left(\frac{d^3y}{dt^3}\right)^2 \right) dt \quad \longrightarrow \quad \frac{d^6x}{dt^6} = 0 \quad \frac{d^6y}{dt^6} = 0$$

$$\longrightarrow \quad \begin{aligned} x(t) &= a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \\ y(t) &= b_0 + b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5 \end{aligned}$$

Add constraints:

$$x(0) = x_0,\ \dot{x}(0) = 0,\ \ddot{x}(0) = 0$$

$$x(T) = x_f,\ \dot{x}(T) = 0,\ \ddot{x}(T) = 0$$

$$y(0) = y_0,\ \dot{y}(0) = 0,\ \ddot{y}(0) = 0$$

$$y(T) = y_f,\ \dot{y}(T) = 0,\ \ddot{y}(T) = 0$$

12 constraints and 12 unknowns

Flash *J Neurosci* 1985 doi:10.1523/JNEUROSCI.05-07-01688.1985

Minimum jerk can determine movement shape

- Minimum jerk
  - Movement should be smooth

$$C = \frac{1}{2}\int_0^T \left(\left(\frac{d^3x}{dt^3} \right)^2 + \left(\frac{d^3y}{dt^3} \right)^2 \right) dt$$

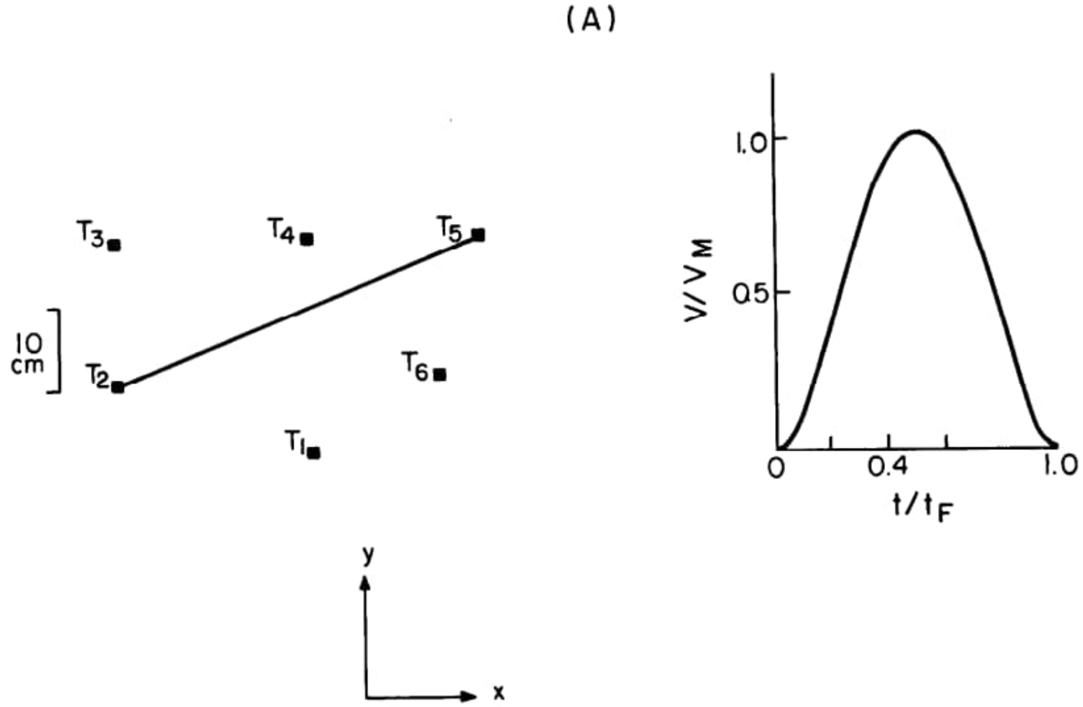
$$x(t) = x_0 + (x_0 - x_f)(15\tau^4 - 6\tau^5 - 10\tau^3)$$

$$y(t) = y_0 + (y_0 - y_f)(15\tau^4 - 6\tau^5 - 10\tau^3)$$

$$\tau = \frac{t}{T}$$

Flash *J Neurosci* 1985 doi:10.1523/JNEUROSCI.05-07-01688.1985

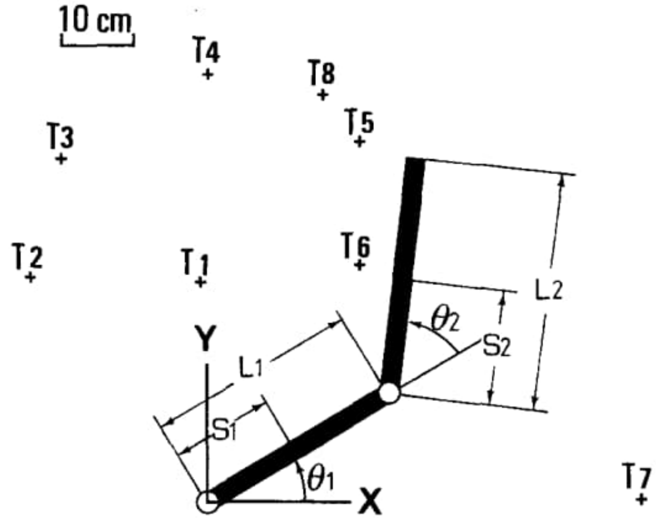
Minimum jerk gives bell shaped trajectory

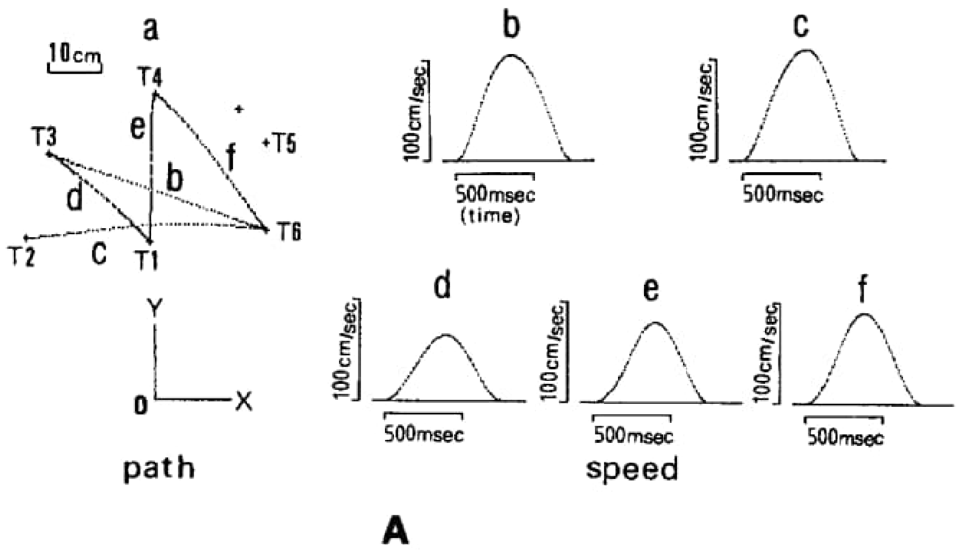


Flash *J Neurosci* 1985 doi:10.1523/JNEUROSCI.05-07-01688.1985

Minimum torque change trajectory

$$C_T = \frac{1}{2}\int_0^{t_f}\sum_{i=1}^{n}\left(\frac{d\tau_i}{dt}\right)^2 dt$$



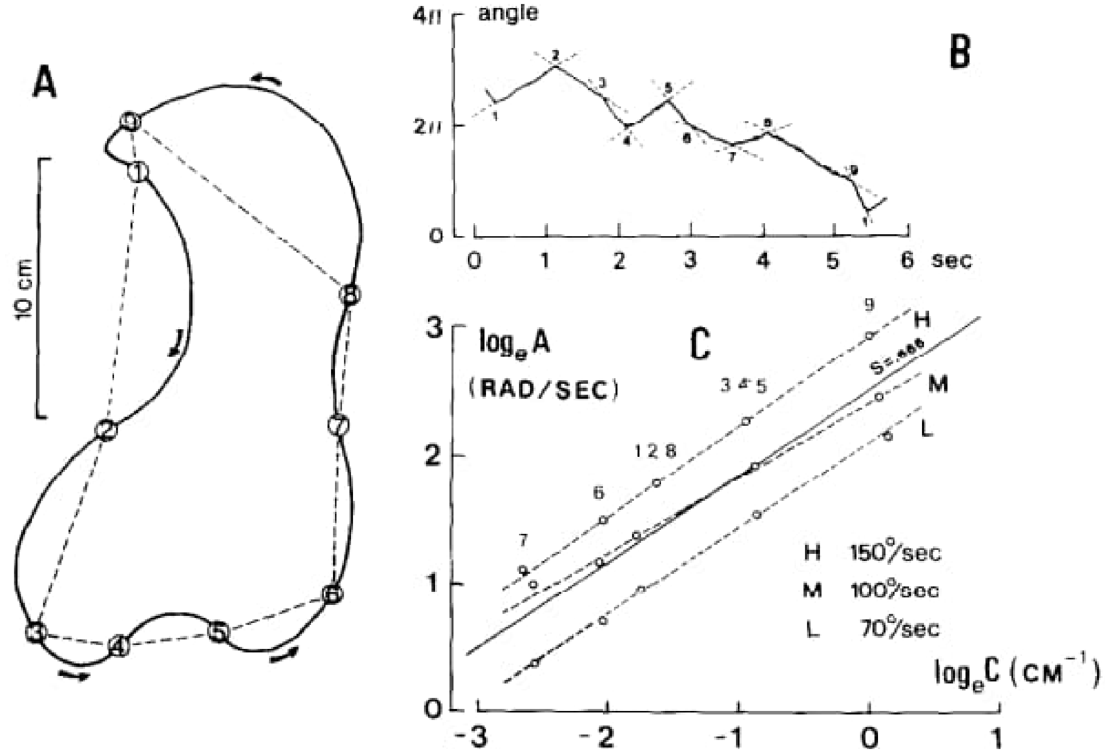


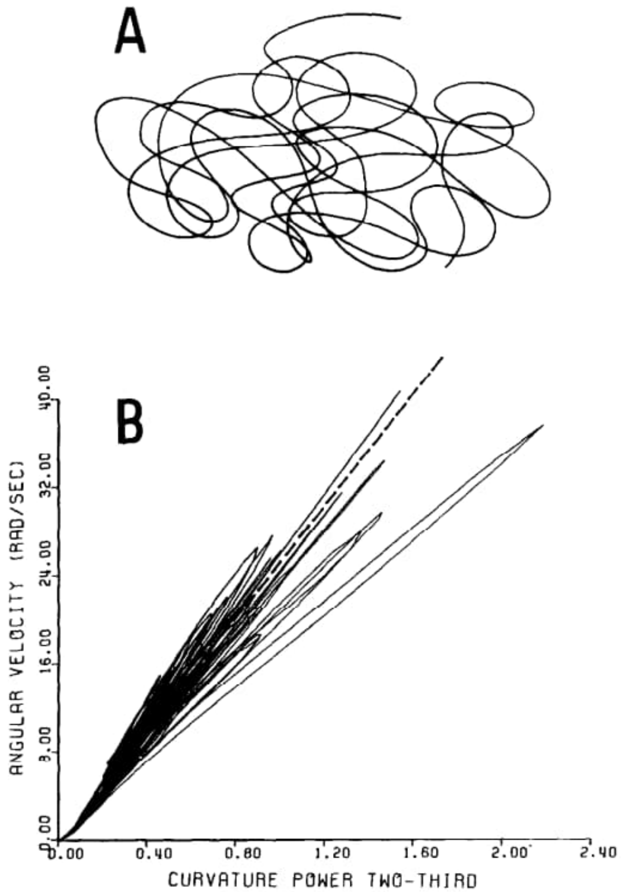
Uno *Biological Cybernetics* 1989 doi:10.1007/BF00204593

Part 5

- The 2/3 power law

2/3 power law

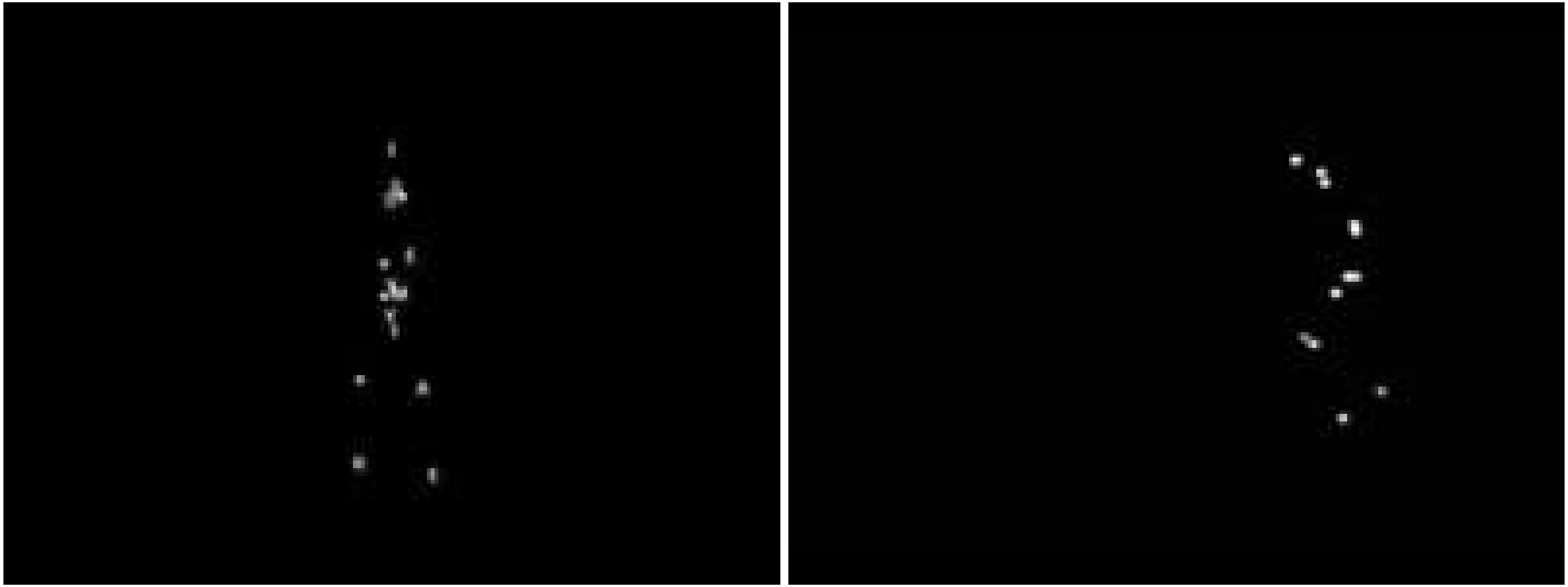




$$A = KC^{\frac{2}{3}}$$

Lacquiniti *Acta Psychologica* 1983 doi:10.1016/0001-6918(83)90027-6

Perception of biological motion

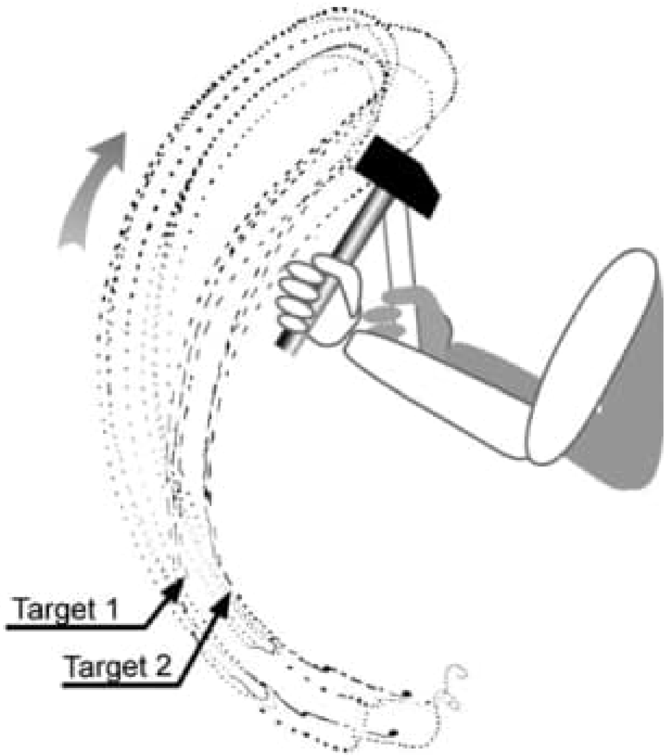


Part 6

- The uncontrolled manifold hypothesis

Not all variability is a problem

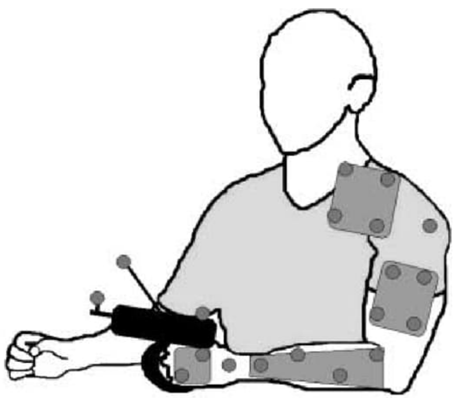
- Down stroke accurate
- Return stroke variable

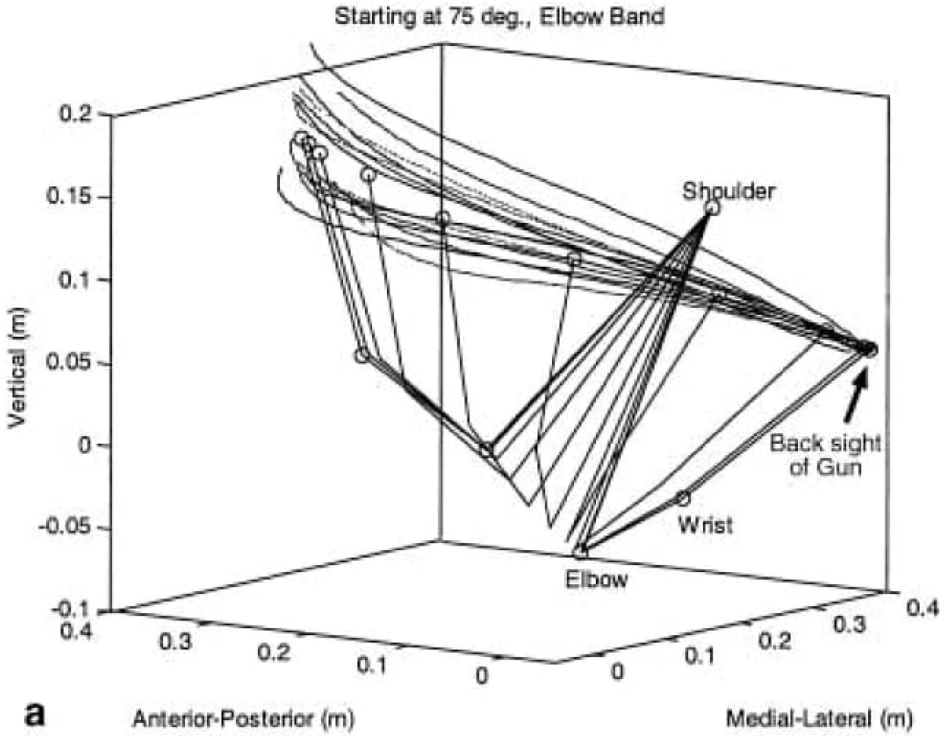


Bernstein 1967 ISBN 9780080119403
via Muller *Advances Exp Med Biol* 2009 doi:10.1007/978-0-387-77064-2_23

The uncontrolled manifold hypothesis

- The nervous does not control variability in dimensions that are not relevant to the task





Sholz *EBR* 2000 doi:10.1007/s002210000540

In class exercise

- Go to Fitt's law and Hick's law online demos.
- Collect and save the data
- Analyze it
  - Scaffolding and step by step instructions in the colab